

GHANA COMMUNICATION TECHNOLOGY UNIVERSITY



FACULTY OF COMPUTING AND INFORMATION SYSTEMS

DEPARTMENT OF INFORMATION TECHNOLOGY

DIPLOMA (INFORMATION TECHNOLOGY) L100 MORNING

END-OF-FIRST-SEMESTER EXAMINATION, APRIL 2025

CIIT 111: INTRODUCTION TO STRUCTURED PROGRAM DESIGN

Duration: 3 Hours

INDEX NUMBER:

Instructions to students:

1. Be sure to fill in your index number on both the question paper and answer booklet before starting the examination.
2. Answer *all questions* in *SECTION A*, *SECTION B*, *SECTION C*, and *only one question* in *SECTION D*.
3. Read all questions and possible answers *carefully*.
4. For multiple-choice questions, tick or circle *only one* letter indicating your answer. Ambiguous selections will be marked wrong!
5. *SECTION A*, *SECTION B*, and *SECTION C* must be answered on the question paper, while questions in *SECTION D* must be answered in the answer booklet provided.

SECTION A (40 MARKS)

ANSWER *ALL QUESTIONS* IN THIS SECTION

1. The position number contained within a square bracket, e.g., my_array [5], is referred to as?
 - A. Post-scripts
 - B. Element of an array
 - C. Sub-script
 - D. None of them
2. This float a [3] = {10.2, 33.4, 44.4}; is an example of _____?
 - A. Initializing variables
 - B. Initializing an array
 - C. Initializing functions
 - D. Initializing float numbers
3. Which of the following functions is defined in <cmath> library?
 - A. pow (a, b)
 - B. power (a, b)
 - C. pow (b)a
 - D. pow(a)_b
4. Machine programming languages are?
 - A. Machine dependent
 - B. Machine independent
 - C. Input dependent
 - D. Output dependent
5. What is the size of the int data type?
 - A. 4 bytes
 - B. 1 byte
 - C. 8 bytes
 - D. 2 bytes
6. What is the output of the program code part written on the console:

```
for (i =2; i <5; i++)  
    cout << i;
```

 - A. 1 2 3 4 5
 - B. 2 3 4 5
 - C. 2 3 4
 - D. 2 4
7. Which among the following operators cannot be overloaded?
 - A. -
 - B. +
 - C. ?:
 - D. %
8. What is std in C++?
 - A. std is a standard class in C++.
 - B. std is a standard file reading header
 - C. std is a standard file writing
 - D. std is a standard namespace in C++.
9. Which of the following will give the size or type of object?
 - A. Size off
 - B. malloc
 - C. realloc
 - D. calloc
10. In C++, you must end a class declaration with what character?
 - A. A right brace.
 - B. Double colons.
 - C. A semi-colon.
 - D. A right curly bracket.

For the next questions from (11) to (15), choose the letter that shows the value that would be in **X**. Assume **X** and **Y** are declared as integer variables. Choose 'error' if you think a code fragment will not compile and run. The code fragments are unrelated (i.e., each question is separate from the others).

11. `int X = 7 - 3 * 2 - 7;`

`X++;`

`cout << X;`

A. 20

C. -20

B. -21

D. Error

12. `X = 2 + 3 || 2 < 3;`

`cout << X;`

A. 5

C. 1

B. 0

D. Error

13. `Y = (30 == 3) && (30 < 40);`

`X = ! Y;`

`cout << X;`

A. 1

C. 27

B. 0

D. Error

14. `Y = 0 >= 0;`

`X = ! Y || -2 <= 0;`

`cout << X;`

A. 1

C. 0

B. -1

D. Error

15. `X = 100;`

`if (3 > 10 || 8 > 14);`

`X = --X;`

`else`

`X = ++ X;`

`cout << X;`

A. 100

B. 101

- C. 99
D. Error
16. A group of _____ can represent each number in hex.
A. 4-bits
B. 8-bits
C. 16-bits
D. 32-bits
17. What is the binary representation 1001001101011010 in hex?
A. 49BB
B. F19B
C. 49AA
D. 40AA
18. An error that causes the program to terminate immediately without having successfully performed its job is called _____?
A. Terminating error
B. Debugging error
C. Syntax error
D. Both A and B
19. Syntax errors are called?
A. Run time error
B. Compilation error
C. Compile time error
D. Both B and C
20. Which language is directly understandable by the computer machine?
A. Assembly language
B. Machine language
C. High-level language
D. All of them
21. _____ is a program in C++ that inputs data from the keyboard or outputs data to the console.
A. stdio file
B. fstream file
C. iostream file
D. none of the above
22. Every variable stored in the computer's memory has a _____.
A. Name
B. Value
C. Size and type
D. All of them
23. A computer is comprised of various devices such as keyboard, mouse, disks, and secondary memory, which are together called _____.
A. Peripherals
B. Software
C. Programs
D. Functions
24. The translator program that converts the assembly language into the machine language is called _____.
A. Assembly compiler
B. Program changer
C. Assembler
D. Pseudo compiler
25. What is the output of the following code?

```
#include <iostream>
#include <string>
using namespace std;
main () {
```

```

int a [ ] = {10, 20, 30};
    cout<<a [2] + a [0] * 30 - a [1];
}

```

- | | |
|--------|--------|
| A. 40 | C. 600 |
| B. 400 | D. 10 |

26. What will be the output of the following C++ code?

```

#include<iostream>
#include<any>
using namespace std;
int main ()
{
    float val = 5.5;
    any var (val);
    cout << var << endl;
    char c = 'a';
    var.emplace < char > (c);
    cout <<var << endl;
    return 0;
}

```

- | | |
|----------|---------|
| A. Error | C. a |
| B. 5.5 | D. 5.5a |

27. What does the next program part write:

```

for (i=2; i<7; i++)
    cout << i;

```

- | | |
|------------------|------------|
| A. 1 2 3 4 5 6 7 | C. 2 3 4 5 |
| B. 2 3 4 5 6 7 | D. 2 4 5 |

28. An error like dividing a number by zero is called a/an _____?

- | | |
|-------------------------|----------------|
| A. Run time error | C. Fatal error |
| B. Execution time error | D. All of them |

29. A consecutive group of memory locations that all have the same name and the same data type is called _____.

- | | |
|---------------|--------------|
| A. Structures | C. Arrays |
| B. Classes | D. Functions |

30. A character array can be initialized using _____.
- A. Float values
 - B. Integer values
 - C. A string literal
 - D. None of them
31. Using the _____ function, we can see the number of bytes taken by a data type.
- A. `sizeof()`
 - B. `sizeof()`
 - C. `sizeof()`
 - D. `sizeof()`
32. The index of the first element in an array is _____?
- A. 1
 - B. 0
 - C. i
 - D. None of them
33. The C++ code `cin >> number` _____
- A. Read from the console.
 - B. Write to the console.
 - C. Read from the variable.
 - D. Assign a variable
34. Using the _____ keyword, we can let the computer into the type of a variable based on its initial value.
- A. `auto`
 - B. `char`
 - C. `sizeof`
 - D. `overflow`
35. The logical _____ reverses a Boolean value.
- A. NOT operator (!)
 - B. AND operation
 - C. OR operation
 - D. None of them
36. We use _____ to repeat the execution of a set of statements.
- A. Variable
 - B. ROM
 - C. Loops
 - D. None of them
37. A typical example of a programming design paradigm is _____.
- A. Unstructured programming
 - B. Structured programming
 - C. High-level programming
 - D. Low-level programming
38. Which memory is closest to the CPU _____.
- A. RAM
 - B. ROM
 - C. Registers
 - D. None of them
39. _____ variable is accessible by all functions.
- A. Local
 - B. Global
 - C. Both A and B
 - D. None of them
40. All are benefits derived from designing a structured program except.

A. Reduced complexity.

C. Easy to package.

B. Easier debugging.

D. Improved readability.

TOTAL (40 MARKS)

SECTION B (20 MARKS)

ANSWER ALL QUESTIONS BY INDICATING WHETHER THE STATEMENT IS

(A) TRUE OR (B) FALSE

1. In writing a C++ program, all local variables are accessible to all functions. _____
2. To call a function in a C++ program, we type its name, followed by parenthesis, and the arguments (if any). _____
3. While and do-while loops are often used when we know ahead of time how many times we need to repeat executing a statement or block of code. _____
4. We use switch statements to compare a variable against different values. _____
5. An if statement can optionally have an else clause in a C++ program. _____
6. With the logical AND operator (&&), if one operand is false, the result is always true.

7. A Boolean expression in a piece of C++ code must always produce a Boolean value.

8. We use the Stream Insertion Operator (>>) to write data to a stream. _____
9. To run a C++ application, we first have to compile our C++ code to the hex code.

10. The main () function is the starting point of a C++ program. _____
11. The correct statement for reading from the console is `cout << "X" << X;` _____
12. The last phase of computer program development is documentation. _____

13. Debugging is translating the program from the source code to the machine code.

14. A computer program written in high-level language requires the knowledge of the computer on which the program is to run. _____

15. An advantage of assembly language programs is that they allow complex tasks to run more simply. _____

16. Variables are always to be initialized before they are used in a C++ program.

17. Low-level computer programming languages are independent of the machine on which they run. _____

18. Comparison operators should always produce integer values. _____

19. Loops based on ranges are helpful when iterating over elements in an array. _____

20. A typical example of a structured computer programming language is Cobol.

TOTAL (20 MARKS)

SECTION C (20 MARKS)

ANSWER *ALL QUESTIONS* IN THIS SECTION ON THE QUESTION PAPER

Q1. State *three (3)* benefits derived from designing a structured program.

[3 marks]

Q2. Briefly state *four (4)* disadvantages of the High-level computer programming languages.

[4 marks]

Q3. Give *four (4)* characteristic features of a good written computer program.

[4 marks]

Q4. List the *four (4)* types of loops that can be implemented in a C++ program.

[4 marks]

Q5. Give *five (5)* examples of built-in data types for storing data in C++ programming.

[5 marks]

TOTAL (20 MARKS)

SECTION D (20 MARKS)

ANSWER ONLY *ALL ONE QUESTION* IN THIS SECTION

QUESTION ONE:

Answer question one using Table I, which indicates the Body Mass Index (BMI) categorization for a user given their calculated BMI value.

TABLE I

BMI Value	Category
< 18.5	Underweight
Between 18.5 and 24.9	Normal weight
Between 25 and 29.9	Overweight
Between 30 and 34.9	Obese Level I
Between 35 and 39.9	Obese Level II
>= 40	Extremely Obese

- A. Write a snippet of C++ program code to allow a user to input their weight(W) in kilograms(kg) and height (H) in meters(m) and display the calculated BMI value using the following formula:

$$\text{BMI} = W \text{ (kg)} / (H \text{ (m)})^2$$

- B. Given the *BMI value of a user*, write a snippet of C++ program code to display the category of the user on the console using *nested if statements*.

TOTAL (20 MARKS)

QUESTION TWO:

Given the 3 X 3 square Matrix below

$$MATRIX = \begin{bmatrix} -4 & 7 & 1 \\ 11 & -3 & 23 \\ 15 & -19 & 2 \end{bmatrix}$$

Write a snippet of C++ program code to do the following:

- A. Store the integer values in the matrix in the computer memory using an array.
- B. Exchange the integer values in the first column with integer values in the last column using a *for loop*.

TOTAL (20 MARKS)

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